

Application Status and Improvement Suggestions for GenAI in China's Education Sector

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Abstract

Generative artificial intelligence (GenAI) has been deeply integrated into all aspects of the higher education in China. Its extensive application reconstructs students' learning experiences through personalized learning protocols and interactive inquiry, empowers teachers in lesson preparation, classroom interaction, and post-class assessment, and promotes intelligent campus administration, balanced resource allocation, and scientific decision-making. However, its application still faces three major challenges: technically AI hallucinations leading to content distortion, insufficient AI literacy among teachers and the degradation of thinking abilities caused by students' over-reliance, as well as ethically academic misconduct and privacy leakage issues. To this end, we need to construct a tripartite collaborative pathway of improvement: at the technical level, enhancing reliability through fact-checking mechanisms and hierarchical filtering; in the model development, creating specialized large models for education to strengthen teaching scenario adaptability and core competency cultivation; in the institutional construction, it is necessary to improve application norms, dynamic supervision, and ethics education, collectively building a safe, efficient, and equitable AI education application ecosystem.

CCS Concepts

• B7;Applied computing; • Education; Computer-assisted instruction.;

Keywords

Generative Artificial Intelligence, Education, Application Status, Suggestions on Improvement

ACM Reference Format:

Yuying Shi. 2025. Application Status and Improvement Suggestions for GenAI in China's Education Sector. In *2025 4th International Conference on Artificial Intelligence and Education (ICAIE 2025)*, November 21–23, 2025, Nanjing, China. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3797552.3797794>

1 Introduction

With the deepening of the new round of technological revolution and industrial transformation, generative artificial intelligence

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ACM ISBN 979-8-4007-2092-5/2025/11

<https://doi.org/10.1145/3797552.3797794>

(GenAI) is integrating into various industries at an unprecedented pace, and the field of education is no exception. In China, GenAI is driving profound changes in the education ecosystem, with its applications expanding from auxiliary teaching tools to comprehensive educational scenarios. In the basic education, intelligent lesson preparation systems provide teachers with personalized instructional designs, and automated essay grading tools offer instant feedback. In the higher education, virtual laboratories break through spatiotemporal constraints, and research assistants accelerate knowledge discovery. In the lifelong learning, adaptive learning platforms dynamically adjust content based on learner characteristics, while AI tutors provide 24-hour query support. These applications have significantly enhanced teaching efficiency and learning experiences, promoting the inclusive sharing of educational resources. However, rapid technological advancement has also introduced a series of urgent challenges: algorithmic bias may lead to educational inequity, data privacy protection remains a concern, the boundaries of human-AI collaboration in teaching are still unclear, the digital divide issue is becoming more pronounced, and there is a lack of specialized models deeply integrated with educational scenarios. Therefore, systematically analyzing the current application status of GenAI in China's education sector, thoroughly studying the theoretical and practical issues it faces, and proposing improvement recommendations suited to China's national conditions are not only of significant practical importance for promoting the digital transformation of education but also provide necessary references for constructing a new educational ecosystem in the intelligent era.

2 The Performance of GenAI in China's Education Sector

Currently, with the smart teaching research scenarios of DeepSeek + University being constructed in China, GenAI has been integrated, at an overwhelming speed, into all aspects of the higher education in China (Figure 1), thus generating unprecedented changes and vitality to traditional teaching models.

2.1 Promoting students' learning

In July 2025, the 56th *Statistical Report on China's Internet Development Status* released by the China Internet Network Information Center (CNNIC) showed that students are the most active group in making the use of GenAI (Figure 2). In September 2025, China Youth Daily, together with Youth Media Alliance and the social platform Soul App, jointly launched a national survey on college students' AI usage behaviors. The newly-released College Students'

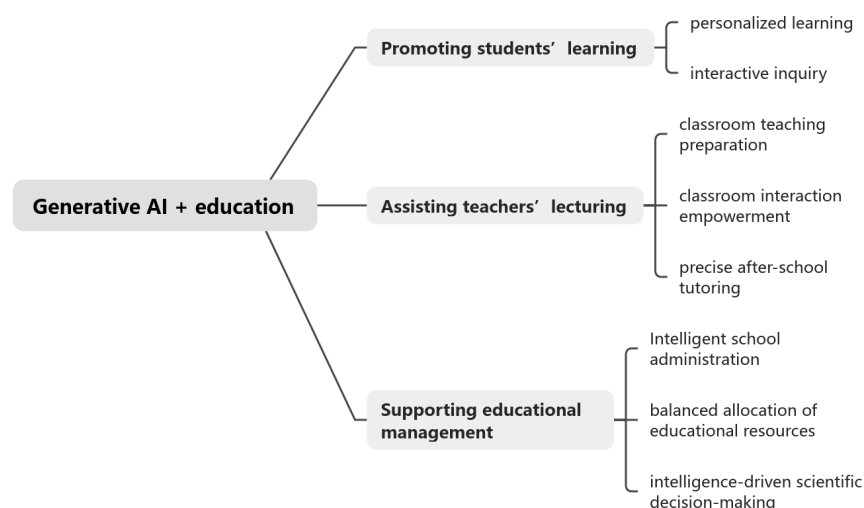


Figure 1: Specific applications of GenAI in China's sector

AI Usage Mindset Insight Report in 2025 showed that 99.2% of respondents had already made use of AI, and 65.9% reported turning to AI as their first response when encountering problems. GenAI has become deeply embedded in students' learning processes, offering diversified and personalized developmental support. This empowerment is mainly manifested in the following two dimensions:

On the one hand, GenAI empowerment supports personalized learning^[1]. Students can leverage GenAI tools to plan and manage their learning progress, receiving tailored learning strategies in the real time. For instance, AI learning companions can generate comprehensive diagnostic reports that assess students' knowledge mastery, cognitive styles, and developmental stages, enabling them to adjust their study pace independently. Through dynamic recommendation systems, AI provides tiered learning materials, adaptive question banks, and enrichment tasks. The real-time feedback continuously refines knowledge graphs, analyzes errors, and formulates targeted improvement strategies, helping students build structured cognitive frameworks. Tsinghua University has launched an AI learning companion system named QingXiaoDa. This system specifically includes a self-assessment feature for academic interests, designed to help students explore themselves at the beginning of their academic journey, discover their unique academic interests and aspirations, and provide targeted course selection advice. Among the surveyed students, over 89% claim it helpful in their feedback. On the other hand, it enables interactive inquiry. GenAI reshapes traditional learning by introducing interactive, immersive experiences via natural language communication and scenarios-based simulations.^[2] In the language learning, for example, students can engage in realistic conversations such as asking for directions on a foreign campus or raising questions at an academic conference. With such immersive activities, students' authenticity and effectiveness in learning can be significantly enhanced.

2.2 Assisting teachers' lecturing

GenAI, with its advantages of efficient processing of multimodal information and unhindered human-computer interaction, is being gradually applied in teaching scenarios such as lesson preparation, classroom implementation, and after-class tutoring. According to the AI Application Report among Teachers in 2025 jointly released by Xueke.com and the Chinese Society for Education Development Strategy, 75% of surveyed teachers had used AI tools in their teaching. Specifically, at the lesson preparation stage, teachers use GenAI to assist in creating teaching designs and courseware. For example, Tianjin No. 2 Xinhua Middle School has introduced and applied the Spark Teacher Assistant, a teaching aid tool developed by iFLYTEK, which can instantly generate complete teaching plans that include teaching objectives, key and difficult points, and extended reading materials. It can also recommend complementary video resources, thus improving the teaching design efficiency by over 56% and courseware creation efficiency by over 64% for teachers. When entering the classroom teaching, a large proportion of teachers will use GenAI tools to conduct interactive teaching and create an immersive teaching atmosphere to improve the effectiveness of classroom teaching. For example, The School of Nursing at Yongzhou Vocational and Technical College has actively introduced cutting-edge AI tools such as ChatGPT and Doubao, integrating them with AI patient simulation technology. This initiative is driving the evolution of the teaching model from a simple teacher-student interaction to a tremendously upgraded stage of teacher-machine-student interaction. In addition, teachers can also turn to GenAI to build teaching AI agents, achieving personalized teaching and intelligent tutoring. For instance, Shanghai Normal University has launched the MetaClass system, a large model for teacher education practice and training based on GenAI. Integrating technologies such as large language models, computer vision,

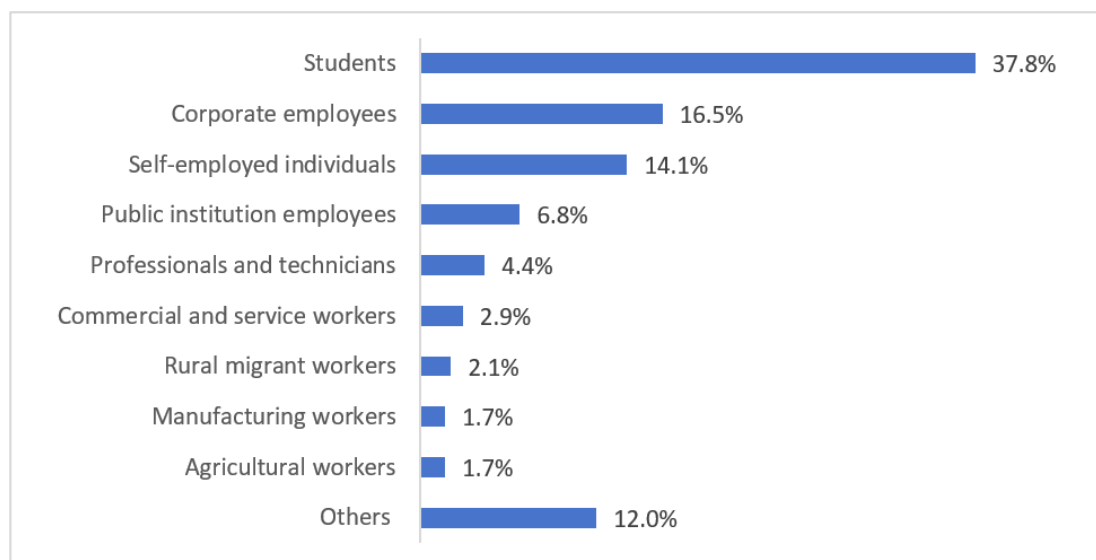


Figure 2: Occupational structure of GenAI users. (Source: Statistical Report on China's Internet Development Status)

digital humans, and human-computer interaction, it builds an immersive human-computer interaction system driven by the large model, thereby creating an intelligent classroom that focuses on human-machine collaboration and interaction-assessment unification. After class, teachers can use GenAI to assist in generating and grading homework, conducting teaching analysis, ability assessment, and effect monitoring, generating matching analysis reports, and recommending precise and optimized after-class tutoring plans. GenAI can also help teachers introduce multi-agent collaborative mechanisms in the teaching evaluation and collaboration phase, conducting multi-dimensional assessments.

2.3 Supporting educational management

GenAI can empower educational administrative departments, schools, and teachers by optimizing daily operations, educational supervision, and decision-making through efficient text generation and data intelligence analysis, thereby significantly improving the accuracy and scientific level of management work. The supporting ways are detailed as follows: Firstly, it promotes intelligent school management. GenAI can be deeply embedded in school administrative processes, optimizing course scheduling through intelligent scheduling systems, reducing repetitive work through automated approval processes, and dynamically integrating and analyzing personnel files, equipment resources, and other data under the premise of ensuring data security and compliance, assisting in formulating resource allocation plans and achieving improved efficiency via cross-departmental collaboration. For example, Northeast University of Finance and Economics has launched Little Ginkgo, a campus intelligent assistant independently developed based on DeepSeek. This system deeply integrates business processes from 13 departments, which has reduced the average handling time by 53% for 387 high-frequency service, such as obtaining academic transcripts and applying for equipment, thus significantly enhancing cross-departmental collaboration efficiency. Secondly, it promotes the

balanced distribution of educational resources. GenAI can convert the high-quality resources such as celebrated teachers' classroom recordings and excellent lesson plans into digital course packages, matching the needs of students in different regions through intelligent recommendation systems, combining virtual laboratories and AI interactive textbooks, effectively breaking geographical limitations so that teachers and students in remote areas can also have an access to the cutting-edge educational resources simultaneously. For example, Beijing University of Posts and Telecommunications, in collaboration with other universities, has developed a 1+N digital education assistance system empowered by GenAI and large model technologies. This system supports flexible expansion of teaching modules, allowing rural primary and secondary schools to independently select required functional components. It enables students to access the on-demand, flexible, personalized, and zero-cost quality educational resources tailored for rural education. The system has been employed in the teaching system of Shexian County, Anhui Province, injecting new vitality into the rural education.

3 Problems Faced with the Application of GenAI in China's Education Sector

Although the prospects are promising, the application of GenAI in China's education sector is still in its infancy, and the accompanying problems cannot be ignored. The problems can be detailed as follows:

3.1 Technical challenges

Currently, the application of GenAI is still faced with challenges of accuracy and reliability. At present, large models often speak nonsense and generate something deviated from facts, a phenomenon known as "AI hallucination". From the technical perspective, there are three principal reasons for the occurrence of AI hallucination. Firstly, insufficient or biased training data leads to defects in the

model's understanding of certain fields; secondly, the limitations of algorithmic architecture, as the mainstream large models are primarily based on probabilistic prediction rather than logical reasoning; thirdly, the setting of training objectives, where models tend to generate seemingly reasonable rather than accurate content.^[3] AI hallucination can lead to significant risks. The Global Risk Report in 2025 released by the World Economic Forum has listed false and misleading information as one of the five global risks, with AI-generated hallucinations being a key factor. As we all know, the accuracy and reliability of information are particularly strict in the education sector. For students whose cognitive systems are not yet mature and lack sufficient discernment, they may easily absorb the erroneous information generated by AI hallucination as authentic knowledge, which not only results in the formation of incorrect knowledge frameworks but also makes it difficult to correct these errors later. At the same time, these erroneous and redundant information also add many troubles to teachers' work, requiring them to spend a lot of extra time and efforts to verify the authenticity of information.

3.2 Pedagogical challenges

Both teachers and students encounter difficulties when integrating GenAI into learning and teaching.

3.2.1 Teachers are not able to apply GenAI in their teaching. GenAI is a new technology, differing significantly from traditional teaching methods and tools. How to skillfully use GenAI is a major challenge for teachers, especially for some senior ones. On September 26, 2025, Fang Zengquan, the Party Secretary of the School of Journalism and Communication at Beijing Normal University and Director of the Center for Minors' Internet Literacy Research, released the Report on Cultivating AI Literacy among Adolescents, which pointed out that most primary and secondary school teachers graduated from non-AI-related majors, and their knowledge, skills, and pedagogical expertise in AI are seriously deficient. The training system for in-service teachers is incomplete, narrow in coverage, and outdated, failing to meet the needs of teaching. The quality and quantity of the teaching staff have become the most critical obstacle to the promotion of AI education in China.

3.2.2 Students are excessively dependent on GenAI. Students' excessive dependence affects the development of their creative thinking, critical thinking, and independent thinking abilities. In February 2025, Xinhua News Agency collaborated with a middle school in the northern region to conduct a questionnaire survey. The results showed that 40% of students had used DeepSeek, Doubao, Kimi, and other AI tools when completing their winter vacation assignments. The MyCOS' Research on AI Application and Literacy among University Faculty and Students in 2025 revealed that although GenAI can assist students in learning, negative impacts also exist. The issues of decreased self-directed learning ability and diminished learning quality resulting from technological dependence are particularly serious (Table 1). According to the cognitive load theory, an appropriate amount of cognitive load is indispensable for the development of thinking. Although GenAI reduces students' extraneous cognitive load by directly providing answers, it also deprives them of the essential intrinsic cognitive load required for integrating

information and constructing schemas during the problem-solving process. This will inevitably leads to the simplified thinking process and hinders the formation of deep cognitive structures. As the learning theory of constructivism maintains, knowledge is not passively received but actively constructed by learners through interaction with problem situations, therefore, over-reliance on GenAI will make students regress from active meaning-constructors to passive information recipients. Ultimately, it weakens the students' ability to internalize and transfer knowledge. In July 2024, an article titled *GenAI Harms Learning* published by the Wharton School Research Paper, stated that AI tools like ChatGPT may temporarily improve students' math test scores, but ultimately weaken their learning abilities. As time goes by, students will be unwilling to think, and their thinking and problem-solving abilities will be seriously weakened.

3.3 Ethical challenges

Some teachers and students utilize GenAI to ghostwrite papers or fabricate experimental data, leading to increasingly serious issues such as academic plagiarism and cheating, which pose severe challenges to academic integrity. In September 2023, the journal *Physica Scripta* withdrew a physics paper, marking the first SCI paper retraction due to ChatGPT usage. The paper was found to contain ChatGPT-specific prompt phrases like "Regenerate response", indicating that the author had directly copied AI output without review or modification. In February 2024, the journal *Frontiers in Cell and Developmental Biology* retracted a paper on spermatogonial stem cells because its illustrations were generated by Midjourney and contained obvious errors—the experimental rats were depicted with four testicles, and the labels included nonsensical terminology. This paper was retracted just three days after publication, setting a record for the shortest-lived paper. There are many other similar cases. In 2025, a study titled *Research on AI Application and Literacy among University Faculty and Students in 2025*, released by the Chinese research institution MyCOS, revealed that 42% of respondents had personally experienced academic disputes triggered by AI-generated content, including plagiarism rulings and assignment deductions. To prevent academic misconducts, many Chinese universities and research institutions have adopted AI detection tools. However, these tools primarily identify AI-generated content through text feature model comparisons and probability threshold determinations, which are subject to technical flaws such as misclassifying original work or failing to detect paraphrased plagiarism. This may undermine genuine academic efforts, incentivize more covert and traceable academic violations, or encourage further opportunistic behavior.

Additionally, GenAI carries the risk of privacy breaches. Student learning records, teachers' instructional materials, and research project information may contain data that should not be publicly disclosed. When such critical data are aggregated and fed into AI systems, the lack of strict compliance reviews and usage protocols could lead to leaks or misuse, potentially harming the rights and interests of teachers and students.^[3]

Table 1: A survey of impact of GenAI on college students' learning

Options	Percentage
Helping better understand and master knowledge	51.1%
Helping broaden learning approaches and horizons	48.5%
Helping enhance learning initiative and motivation	43.2%
Helping improve learning efficiency and saved time	42.7%
Having caused a certain level of technological dependence and a decline in self-directed learning ability	37.3%
Having resulted in a decline in learning quality (like insufficient depth in assignments or improper use of AI-generated content)	24.4%
Having shown no discernible impact on learning	0.6%

(Source: MyCOS' Research on AI Application and Literacy among University Faculty and Students in 2025)

4 Improvement Suggestions for the Application of GenAI in China's Education Sector

Faced with the challenges in the application of GenAI in the field of education, all parties need to work together to build a safe, efficient, and fair AI education application ecosystem through technological improvements, model optimization, and institutional construction.

4.1 Promoting technological improvements and innovations

The technology of GenAI is still in the burgeoning stage of development, so it is necessary to continuously break through technical bottlenecks to enhance its applicability, safety, and reliability in China's education sector. To achieve this goal, the following measures should be taken:

4.1.1 The training objectives and evaluation system of AI systems should be optimized. Currently, some GenAI models excessively prioritize the fluency and naturalness of generated text while neglecting the core requirements of factual accuracy. To address this, it is necessary to reconstruct training objectives from a technical perspective, treating fact-checking capability as an equally important evaluation dimension as expressive ability.^[4] The specific implementation pathways include: anchoring training data to authoritative educational resources, prioritizing the use of textbooks reviewed by educational authorities and resources from national platforms like the Smart Education Platform for Primary and Secondary Schools as training data to replace the web texts, thereby ensuring knowledge accuracy at the source; designing an evaluation index system that includes multiple dimensions such as factual accuracy, logical rigor, and expressive fluency. Based on the national curriculum standards, this system quantitatively assesses the consistency of AI-generated content with textbooks and authoritative resources. For example, it evaluates whether the AI's description of the historical significance of the 1911 Revolution covers the key textbook points such as overthrowing the feudal monarchy and embedding the concepts of democracy and republicanism deeply in people's minds, with corresponding points deducted for non-compliance.

4.1.2 Credible traceability and cross-verification mechanisms should be constructed. For AI-generated educational content, it should be

mandatory to annotate the information sources. For instance, methods such as embedded links or hover tooltips should be used to clearly indicate the authoritative academic databases, approved textbooks, or core journal literature as data sources, making information traceability an operable and verifiable learning component. For the key knowledge points involving scientific principles, historical events, etc., the system must automatically invoke multiple credible data sources for cross-verification. For example, when explaining historical events, the system can simultaneously retrieve records from the National Archives, authoritative historical research institutions, and multiple authoritative history textbooks to conduct a comparative analysis of descriptions from different sources. When conflicting information exists across sources, the system should immediately alert the user to the points of dispute, annotate the authority ranking of each source, and recommend consulting the latest revised authoritative textbooks or consulting relevant field experts to prevent cognitive errors caused by biases in a single information source.

4.2 Developing education-specific large models

General-purpose GenAI models struggle to fully meet the specific demands of educational scenarios. Developing specialized large models for the education sector has become an inevitable choice to promote the deep integration of AI and education. Educational specialized large models should fully embody pedagogical principles, cognitive characteristics, and teaching needs, providing teachers and students with AI tools that truly understand and are designed for education.^[5]

4.2.1 Key development directions for education-specific large models.

The development of large-scale educational models should prioritize creating specialized models tailored to distinct academic stages and subject-specific needs. For example, a Chinese Writing Large Model must adhere strictly to the national curriculum standards, assisting students in analyzing textual structure and paragraph logic while offering advanced writing support such as emotional expression refinement. Conversely, a Vocational Education Smart Manufacturing Large Model should integrate industry standards to generate standardized training videos and other instructional materials, thereby enhancing teaching clarity and practical relevance. A Teacher-Model Collaborative Interaction Interface is essential to

enable educators to flexibly adapt and refine AI-generated content, for instance, by modifying instructional activities based on classroom dynamics. The model should concurrently learn from such adjustments, progressively personalizing content generation and establishing a feedback-driven cycle between pedagogical practice and algorithmic improvement. Furthermore, the model training should incorporate core competencies such as critical thinking and innovation into its evaluation framework. This ensures that generated content promotes students' holistic development. For instance, when proposing scientific experiments, the model should not only furnish standard procedures but also encourage exploration of diversified experimental variable control strategies, thereby fostering innovative thinking and practical skills.

4.2.2 Feasibility analysis of developing education-specific large models. From China's practical perspective, the development of specialized educational large models faces real-world challenges such as limited data resources, a shortage of interdisciplinary professionals, and substantial funding requirements. It is essential to establish collaborative models that integrate fragmented resources and lower the barriers to research and development. Specifically, efforts should be made to promote a synergistic mechanism involving government, industry, academia, and research institutions. Guided by the national education digitalization strategy, universities, research institutes, and enterprises should jointly establish high-quality educational data alliances and computing infrastructure. Additionally, cross-regional and cross-institutional collaborative R&D models should be explored, encouraging the use of open-source collaboration and shared licensing mechanisms to reduce development costs and enhance resource utilization efficiency. This will gradually build a sustainable and scalable development pathway for educational large models.

4.3 Enhancing teachers' and students' AI literacy and application capabilities

Enhancing both the teachers' and students' understanding and application capabilities of GenAI is a crucial step in deeply integrating this technology into teaching and learning scenarios and advancing the digital transformation of education.

4.3.1 A tiered and progressive AI training system for teachers should be constructed. In response to the differences in the teaching workforce regarding subject background, technical foundation, and teaching experience, a tiered and categorized training system should be established. This system comprises three interconnected levels. The basic level targets all teaching staff, focusing on AI fundamentals and their alignment with educational contexts. It covers practical skills such as operating intelligent teaching tools and utilizing educational data platforms, aiming to equip educators with essential competencies for AI-assisted instruction. The advanced level is intended for the core teachers with some technical backgrounds, emphasizing the deep integration of AI into subject-specific teaching. Through the formation of interdisciplinary teaching research teams, this tier facilitates the co-development of an "AI+Subject" teaching case library, guiding participants in designing tailored AI-enabled instructional strategies. The innovative level caters to expert teachers with strong exploratory motivation,

aiming to cultivate their capacity to engage with cutting-edge AI educational applications. This can be realized through specialized workshops and research projects led in collaboration with university researchers and industry engineers, focusing on innovative uses of GenAI in instructional design, process evaluation, learning intervention, and related domains.

4.3.2 The AI literacy-based teacher evaluation and incentive mechanism should be improved. To effectively stimulate teachers' intrinsic motivation to learn and apply AI technology, education management departments and schools should collaborate to incorporate AI literacy and application capabilities into the comprehensive teacher evaluation system. Specifically, teachers who proficiently master AI technology in teaching and achieve remarkable results should be given clear policy preferences in aspects such as recognition and awards, professional title evaluation, and performance-based compensation. For example, a dimension for "AI in educational application" could be added to the professional title evaluation criteria, giving priority in promotion to teachers who develop high-quality AI teaching cases, form replicable teaching models, and validate them effectively in practice. Meanwhile, regular selection of "AI in Education Teaching Leaders" could be organized, with special performance-based rewards granted to teachers who demonstrate excellence in AI teaching applications and effectively enhance teaching quality and student learning outcomes. These institutionalized incentive measures are aimed to create strong positive guidance and foster a supportive institutional environment that encourages teachers to proactively learn, actively apply, and boldly innovate.

4.3.3 Students' literacy in the rational use of AI should be cultivated. AI literacy education should be comprehensively and systematically promoted across all educational stages, from primary and secondary schools to universities. Curriculum design should go beyond simple tool introduction and guide students to deeply understand the basic concepts, core functions, and inherent limitations of AI, enabling them to clearly recognize what AI technology can and cannot do. Course content must include an analysis of potential risks associated with AI, such as data privacy and security, algorithmic bias and discrimination, technological dependency, and other social-ethical issues, to foster students' awareness of risk prevention. More crucially, the curriculum system should emphasize the training of students' critical thinking skills. Through teaching activities such as project-based learning and case analysis, students should be cultivated to maintain a cautious and questioning attitude toward AI-generated content, neither readily trusting nor blindly following it, and to master basic methods for verifying information through multiple channels and discerning the authenticity of content. The ultimate goal is to nurture them into rational digital citizens who can not only effectively use technology to empower their learning and daily lives but also maintain independent thinking and value judgment.

4.4 Strengthening the construction of relevant systems

The healthy development of GenAI in the education sector relies not only on technological advancement but also on sound institutional safeguards. By improving policies and regulations, industry

standards, and governance mechanisms, an institutional environment for AI applications in education should be constructed, one that stimulates innovative vitality while effectively managing risks.

4.4.1 The application norms and standard systems should be established. Targeting the key processes such as data collection, algorithm training, and content generation, ethical guidelines on privacy protection, information authenticity, and value orientation should be clarified, and operational specifications for data desensitization, content review, and algorithm filing should be established. [6]

4.4.2 The coordinated development mechanisms should be strengthened. The integration of GenAI and education is a systematic project, which requires breaking down barriers among different entities and establishing a collaborative development framework involving the government, enterprises, schools, and research institutions to form a cooperative mechanism characterized by complementary advantages and clearly defined rights and responsibilities. At the government level, it should play the role of a pilot by introducing industrial support policies and establishing special research and development funds to guide social resources toward the field of AI in education. As the technology supplier, enterprises need to strengthen demand coordination with schools, conduct in-depth research at the frontline of education and teaching, and customize AI products according to the teaching needs of different academic stages and subjects, so as to avoid disconnection between technology development and educational practice. Meanwhile, enterprises should also undertake the responsibility of technical training by providing guidance and skill training for teachers and students on the use of AI tools. As the application terminal, schools need to establish internal coordination mechanisms, clarify the division of responsibilities among the academic affairs office, information center, subject teaching research groups, and homeroom teachers, and form a closed loop of technical support–teaching application–student management. At the same time, schools should actively provide feedback on problems and needs encountered in AI applications, offering practical basis for enterprises to optimize technology and for the government to adjust policies. Research institutions should focus on theoretical and technical breakthroughs, conducting research on AI education ethics, application models, evaluation systems, and other aspects, so as to provide academic support for coordinated development.

4.4.3 A dynamic monitoring and evaluation mechanism should be set up. To ensure the sustainability and effectiveness of GenAI applications in education, it is necessary to conduct real-time monitoring and regular evaluations of their implementation process, systematically collect and analyze application outcomes and potential risks, and provide a basis for policy adjustments and practical improvements. In terms of building the monitoring system, efforts should be made to promote the development of an integrated and intelligent monitoring platform that consolidates AI product operation data, user behavior data of teachers and students, teaching process data, and academic achievement data. A multi-dimensional monitoring indicator system covering technical reliability, pedagogical suitability, and ethical compliance should be constructed to enable early identification and warning of risk factors. In terms of designing the

evaluation mechanism, professional institutions should be engaged to develop a scientific evaluation framework. Centered on core objectives such as the improvement of educational quality, the promotion of student development, and the empowerment of teacher capabilities, formative and summative evaluations should be carried out. By integrating multi-source data, a comprehensive analysis of the short-term effects and long-term impacts of AI applications in education should be conducted, forming a closed-loop governance pathway of monitoring–evaluation–feedback–optimization.

5 Conclusion

GenAI is profoundly reshaping the ecosystem of education in China, driving a shift from knowledge transmission to competency development, from standardized teaching to personalized learning, and from experience-based management to data-driven intelligence. This transformation is not merely a process of technologically enhancing existing educational models, but also a deep restructuring of the essence, value, and paradigms of education. In the future, the deeper application of GenAI in the educational field requires coordinated advancement in the following four areas: technological improvement, model development, literacy enhancement, and institutional safeguards. Only in this way can the technical advantages of GenAI be fully leveraged, its deep integration with educational endeavors achieved, and sustained momentum injected into high-quality educational development.

Acknowledgments

The authors have no competing interests to declare that are relevant to the content of this article.

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